

A Step-by-Step Demo: Engaging in the projects using Git

Step 1: Install Git

- Download and install Git from the official website, <https://git-scm.com/downloads> . Follow the instructions for your operating system. i.e Windows, macOS or Linux.
- Remember to choose the default options unless you have specific preferences.
- Verify your installation using this prompt:

```
bash
```

```
1 git --version
```

Step 2: Set Up Git

1. **Configure Your Git Identity by setting up your username and your email**

```
bash
```

```
1 git config --global user.name "Your Name"
```

```
bash
```

```
1 git config --global user.email "your_email@example.com"
```

2. **Check Configuration by running:**

```
bash
```

```
1 git config --list
```

Step 3: Clone the Repository Locally

1. Clone the Repository to create a local copy of the repo: (An Illustration)

- Copy the repository URL (HTTPS or SSH).
- Open your terminal and navigate to the directory where you want to store the project.
- Run the following command (replace <repository-url> with the copied URL):

```
bash

1 git clone <repository-url>
```

Step 4: Add Project Files

1. Navigate to the Repository Directory

```
bash

1 cd <repository-name>
```

Step 5: Open the Repository in VS Code

1. Open the Cloned Repository:

- In VS Code, go to File > Open Folder... and select the folder of the cloned repository.

Step 6: Add or Modify Project Files

1. Edit Files:

- Use the VS Code editor to add new files or modify existing ones. You can create new files by right-clicking in the Explorer pane and selecting New File.

Step 7: Stage, Commit, and Push Changes

1. Stage Your Changes:

- In the terminal, after making changes, stage the files you want to commit:

bash



```
1 git add .
```

2. Commit Your Changes:

- Commit the staged changes with a descriptive message:

bash



```
1 git commit -m "Your commit message here"
```

- Or you can also commit using the Source Control view by entering your commit message in the input box at the top and clicking the checkmark icon.

3. Push Changes to GitHub:

- Push your changes to the remote repository
- Replace main with the name of the branch you are working on.

bash



```
1 git push origin main
```

Step 9: Create a Pull Request (Optional)

1. Go to the Repository on GitHub:

- Click on the "Pull requests" tab.
- Click on "New pull request."

2. Select Your Branch:

- Choose the branch you pushed your changes to and compare it with the main branch.

3. Create the Pull Request:

- After selecting the branches to compare, you will see a summary of the changes.
- Add a title and description for the pull request to explain what changes you made and why.
- Click the "Create pull request" button to submit your pull request.

